

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Low Power VLSI Design [ECE-4063]

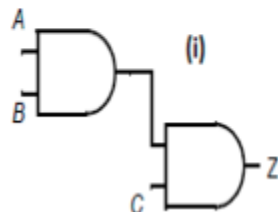
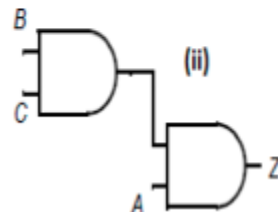
Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

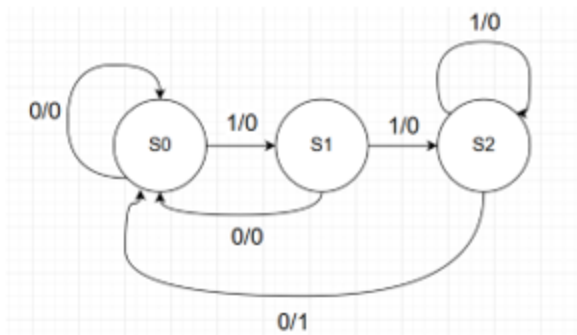
- 1) Compute the transition density and static probability of $Y=ab+c$. Given $P(a)=0.1$, $P(b)=0.2$, $P(c)=0.3$, $D(a)=1$, $D(b)=2$, $D(c)=3$ (4)
 - A)
 - B) Draw ROBDD for $F=A.B.C$. Calculate the output probability for the ROBDD using Thornton and Nair's Top down algorithm (3)
 - C) The chip size of a CPU is $15\text{mm} \times 15\text{mm}$ with clock frequency of 300Mhz operating at 6V . The length of the clock routing is estimated to be twice the circumference of the chip. Assume the clock signal is routed on a metal layer with width of $1.5\text{ }\mu\text{m}$ and the parasitic capacitance of the metal layer is $2\text{fF}/\mu\text{m}^2$. Estimate the power dissipation of the clock signal. (3)
- 2) Implement the following function using Sleep Transistor logic (4)
 - (a) $F=(A.B)'$ (b) $F=(A+B)'$ (c) $F=a+bc$ (d) $F=a'b'+c$
 - A)
 - B) Consider the two structures as shown in figure with $P_A=0.5$, $P_B=0.2$, $P_C=0.1$. Compute the activity factor for both the cases and find the input ordering for minimum power dissipation (3)



 - C) For the given function $Y=AB+C$, Calculate Entropy for all the input, output and intermediate nodes. (3)
- 3) (4)

A)

The state diagram of a Mealy machine for a 110 sequence detector is shown below. The P_{ij} for **S0→ S1 transition** is 0.25 and $P_{ij} = 0.50$ for rest of the cases. Estimate the

Wij (ii) Objective functions (iii) Power dissipation for both assignments. Neglect self transitions/loop.



State	Assignment-1	Assignment-2
S0	00	01
S1	10	10
S2	01	11

B)

TV tower is sending following set of serial data to the base station. It has been observed that a lot of power dissipation is taking place. Calculate the reduce number of transitions, power dissipation using SILENT coding technique on the data sent

Name	T	T+1	T+2	T+3	T+4
D5	1	1	0	0	0
D4	0	1	0	1	0
D3	1	1	1	1	1
D2	1	1	0	1	0
D1	1	0	1	1	1
D0	0	1	0	0	0

(3)

C)

(3)

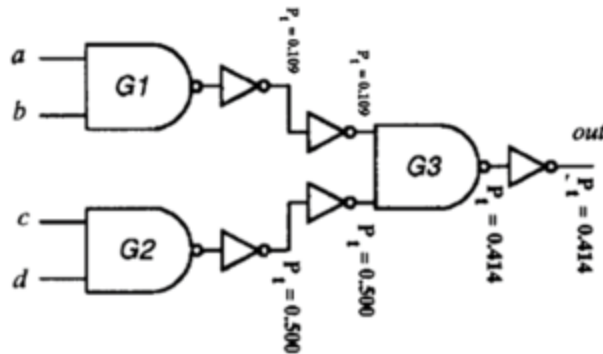
A satellite is sending the following set of important serial data to the base station. It has been observed that a lot of power dissipation is taking place . Calculate the reduce number of transitions, power dissipation using weight based bus invert coding technique on the data sent.

Name	T	T+1	T+2	T+3	T+4
A5	0	0	0	1	1
A4	0	0	1	0	0
A3	0	1	1	1	1
A2	1	1	0	0	0
A1	1	1	1	0	1
A0	0	1	0	0	0

- 4) Calculate the power dissipation of 3 bit binary to gray code sequence. Determine the code has minimum power dissipation (4)
- A)
- B) Consider the two-bit magnitude comparator implemented using pre-computed logic. Analyze the operation of A=1010 and B=0100 magnitude comparator with neat circuit diagram using pre-computed logic block. (3)
- C) 8 bit data "00000000" needs to be sent over a channel. Mr. X wants to switch it to -1 either in 2's complement and signed magnitude form . Compare the two forms and determine which has less power dissipation. (3)
- 5) Estimate the total load capacitance, consider H-tree based clock routing of h=5 levels. (4)
- A) Given the total number of clock terminals $N=4$, the nominal input capacitance at each terminal, $c_g=2$ pF , the unit length wire capacitance, $c_w=0.1$ pF/ μm and the chip dimension, $D=4$ μm , $\alpha=0.1$ μm .
- B)

(3)

In a development of ASIC, following circuit shown in 5b is to be mapped with standard cell library for low power applications . Estimate the optimized mapping so that power dissipation can be reduced. Do calculations for both Trivial case and optimized case.



Standard cell	Area(μm^2)	O/p Cap.(nF)	I/p Cap.(nF)
INV	928	0.1029	0.0514
NAND2	1392	0.1421	0.0747
NAND3	1856	0.1768	0.0868
AOI21	1856	0.2310	0.0850

- C) One hot encoding is used in the sender of the transmitter. At the receiver side, the one-hot code is converted into gray code. Design a hot code to 3 bit gray code converter for low power dissipation. (3)

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